

Homework 1

CHE446 – Fall 2024

Due Friday, August 30th at 11:59 pm

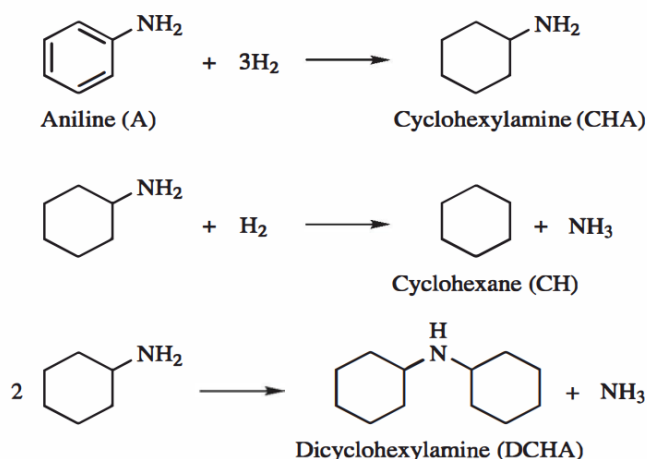
5 problems; 100 points.

NOTE: for problems asking for a “one-page memo:” just work the problem as usual, clearly showing all steps. No formal memo necessary (unless you’re feeling particularly creative).

1. Read and sign the content sharing form (provided on moodle). If hand-writing this assignment, you can read the form, then simply write “Content Sharing Form” and sign your name to represent the form.

2. Problem 1-6:

Problem 1-6 (Level 1) The hydrogenation of aniline at about 50 °C, over a Ru/carbon catalyst, is believed to involve the reactions²:



In one particular experiment, aniline was hydrogenated in a closed vessel at 50 °C and 50 bar of H₂ pressure for 3 h. The following data were obtained:

Species	Moles after 3 h/mol of aniline charged
A	0.476
CHA	0.346
CH	0.080
DCHA	0.049

The amounts of ammonia formed and H₂ consumed were not measured.

1. Is the experimental data consistent with the assumption that these three reactions are the only ones that occur?
2. Estimate the amount of NH₃ formed (mols/mol A charged).
3. Estimate the amount of H₂ consumed (mol/mol A charged).

3. Problem 1-3:

Problem 1-3 (Level 1) The following e-mail is in your in-box at 8 AM on Monday:

To: U. R. Loehmann
From: I. M. DeBosse
Subject: Quinoline Hydrogenation
U.R.,

I hope that you can help with the following:

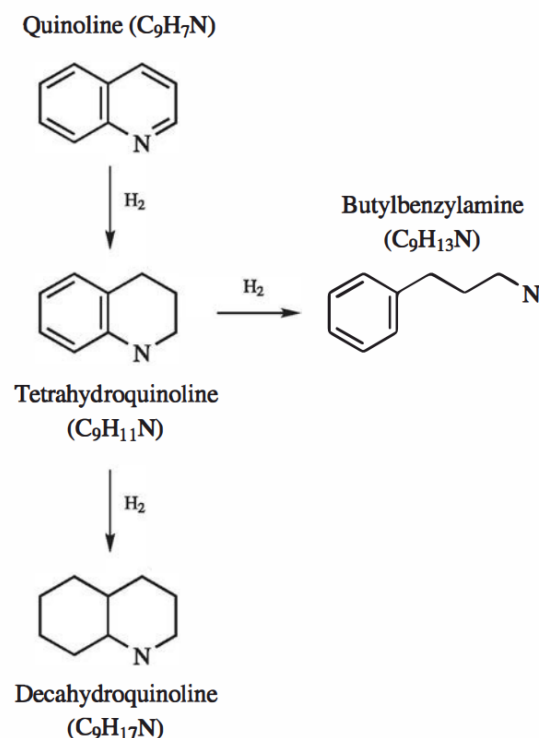
When quinoline (C₉H₇N) is hydrogenated at about 350 °C over various heterogeneous catalysts, the three reactions shown below take place to varying extents. In one experiment in a batch reactor (closed system), the initial charge to the reactor was 100 mol of quinoline and 500 mol of hydrogen (H₂). After 10 h, the reactor contents were analyzed, with the following results:

Quinoline (C₉H₇N)—40 mol

Hydrogen (H₂)—290 mol

Decahydroquinoline (C₉H₁₇N)—20 mol

If the reactions shown below are the only ones that take place, how many moles of tetrahydroquinoline (C₉H₁₁N) and butylbenzylamine (C₉H₁₃N) should have been present after 10 h?

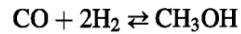


Please write me a short memo (not more than one page) containing the results of your calculations and explaining what you did. Attach your calculations to the memo in case someone wants to review the details.

Thanks, I. M.

4. Problem 1-2: (There are many possible hypotheses; just propose something reasonable)

Problem 1-2 (Level 2) A continuous reactor operating at steady state is being used to study the formation of methanol (CH_3OH) from mixtures of H_2 and CO according to the reaction



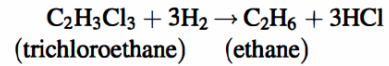
Some data from one particular run are shown below.

Species	Rate in (g·mol/min)	Rate out (g·mol/min)
CO	100	83
H_2	72	38
CO_2	9	9
CH_4	19	19
CH_3OH	2	13

Does the system behave as though only one stoichiometrically simple reaction, i.e., the above reaction, takes place? If not, develop a hypothesis that quantitatively accounts for any discrepancy in the data. How could your hypothesis be tested?

5. Problem 1-8:

Problem 1-8 (K2-1) (Level 1) Consider the reaction



If the rate of formation of HCl (r_{HCl}) is $25 \times 10^{-6} \text{ g·mol/g·cat·min}$

1. What is the rate of disappearance of trichloroethane?
2. What is the rate of formation of ethane?